

ABSTRACT

Predicting NCAA March Madness Outcomes

This project investigates predictive modeling of NCAA Division I basketball tournament outcomes through statistical and machine learning approaches. The goal is to build upon traditional statistical rankings by developing models that generate well calibrated win probabilities for every possible tournament matchup. Following a comprehensive review of existing literature, three modeling frameworks were constructed using R: logistic regression trained on Dean Oliver's Four Factors of Basketball Success, an XGBoost ensemble model incorporating tournament seeding, strength of schedule adjusted box score statistics, and a custom Elo rating system, and a Monte Carlo simulation framework that derives win probabilities from KenPom efficiency based point predictions. All models are evaluated using the Brier Score, which penalizes both incorrect and overconfident probability estimates. The findings indicate that ensemble methods incorporating dynamic rating systems provide a competitive edge in forecasting the high variance environment of single elimination tournament play. The complete codebase is hosted on GitHub for further research and reproducibility: <https://github.com/JonathenMarin/March-Mardness>.

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TABLE OF CONTENTS

	Page
LIST OF TABLES	v
LIST OF FIGURES	vi
INTRODUCTION	1
Data Available	4
LITERATURE REVIEW	10
Past Strategies	10
Modeling Techniques	13
Probability Prediction Models	17
Logistic Regression	18
Boosting	35
Point Spread Models	50
s	50
Comparison of models	51
2026 Competition	52
Strategies	52
Results	52
CONCLUSIONS	53
Summary	53
REFERENCES	54

LIST OF TABLES

	Page
Table 1. Snapshot of raw Kaggle regular-season data demonstrating the 'W' and 'L' column structure.	23
Table 2. Logistic Regression Backtest Results vs. Kaggle Benchmarks	28
Table 3. Logistic Regression Coefficient Estimates (Men's 2025 Model)	29
Table 4. Logistic Regression Coefficient Estimates (Women's 2025 Model)	30

LIST OF FIGURES

	Page
Figure 1. Regular Season Detailed Data	5
Figure 2. The first 20 teams of the KenPom data table.	7
Figure 3. The first 10 teams of the Barttorvik table	8
Figure 4. A sample of the Massey Ordinals table showing distinct ranking systems.	9
Figure 5. An Example of a Logistic Regression Equation [12].	14
Figure 6. An Example of a Linear Regression Equation [13].	14
Figure 7. Confusion Matrices for the 2025 Logistic Regression Models	33
Figure 8. Left: Average point differential by Team 1 seed. Right: Average point differential by seed differential.	38
Figure 9. Left: Top 20 men's programs by end of season Elo rating in 2025. Right: Top 20 women's programs by end of season Elo rating in 2025.	43
Figure 10. Example of Cross Validation and Regular Test/Train Split [19]	48

INTRODUCTION

The National Collegiate Athletic Association (NCAA) serves as the premier developmental system for the National Basketball Association (NBA) and the Women's National Basketball Association (WNBA). With thousands of athletes across both divisions arriving from around the world, the influence of the NCAA is extensive. As the basketball world has evolved into a multibillion dollar industry [1], the value of a competitive edge has become substantial. Teams historically relied on live scouting and film study to gain these advantages. However, the last decade has seen a major shift toward statistical findings. Inspired largely by the movie *Moneyball*, teams now look to analytics to uncover value that the naked eye might miss.

The focal point of this competitive landscape is March Madness. At the conclusion of the regular season, the selection committee invites 68 teams to participate in this tournament. The field includes the champions from all 32 Division I conferences alongside the best remaining teams chosen at large. The committee seeds these participants into four distinct regions with rankings ranging from 1 to 16. The four teams judged to be the strongest are awarded the number 1 seeds while the next tier receives the number 2 seeds. Within each region, the teams compete in a single elimination format where matchups are initially determined by rank. For example, the top seed plays the sixteenth seed. The winner of each region advances to the Final Four to compete for the national championship. Crucially, these contests take place at neutral sites rather than on home courts. This structure creates a predictive challenge where volatility is the norm.

To evaluate these predictive challenges in a rigorous and competitive environment, this project utilizes the Kaggle March Machine Learning Mania competition as its primary experimental framework. Kaggle provides a unique platform where data scientists from across the globe compete to minimize error in tournament forecasts using historical data stretching back to 1985. Unlike traditional bracket challenges that reward binary picks, this competition requires the submission of win probabilities for every possible matchup in the tournament field. This shift from simple selection to probabilistic estimation allows for a more granular analysis of model performance and calibration. By using this structured setting, the research can leverage standardized datasets including team efficiency ratings and historical results to build a pipeline that is both reproducible and statistically sound.

This statistical consistency is possible because basketball is uniquely suited for machine learning applications. Both individual and collective performances are captured through a standardized set of metrics. Discrete variables such as points, steals, assists, and rebounds provide a consistent numerical foundation for analysis across different eras and levels of competition. This uniformity allows for the creation of structured datasets where team strength can be quantified and compared through objective performance indicators. By leveraging these recurring statistical patterns, machine learning models can identify underlying trends that govern game outcomes and team efficiency.

A basic assessment often relies on a record of wins and losses to infer the overall quality of a team. While these records serve as strong indicators,

they frequently fail to capture critical nuance. Furthermore, the single elimination structure of the tournament ensures that upsets remain a constant possibility regardless of historical dominance. A defining example occurred in when the top ranked Virginia Cavaliers entered the postseason with 31 wins and only 3 losses. They fell to the lowest ranked UMBC Retrievers by 20 points. Such outcomes demonstrate why simple heuristics fail in this environment.

This inherent volatility drives a massive secondary market focused on sports betting. As legal wagering expands globally, March Madness has evolved into one of the largest gambling events in the world with 3.1 billion dollars wagered in 2025 on tournament outcomes [2]. To manage this risk, bookmakers generate odds which serve as the industry standard for predictive accuracy. These odds represent a market consensus that attempts to price in every available variable from player injuries to historical trends. However, these lines are often influenced by public sentiment and profit margins rather than pure probability. Consequently, developing a model capable of outperforming these market established baselines represents the gold standard for predictive success.

This paper aims to construct a predictive framework capable of meeting this standard. The primary objective is to develop and evaluate distinct statistical models that minimize the Brier Score metric used in the Kaggle competition. By testing various algorithms against historical data, the research seeks to identify a strategy that generates a winning score. The final analysis benchmarks these models against competitor submissions that utilized multiple strategies including elo ratings, standard odds conversion,

and analysis rankings. Ultimately, the optimal model will be deployed in the upcoming 2026 March Madness tournament to compare its projections with live results.

Motivated by this challenge, substantial research has addressed the complexities of March Madness prediction. The majority of this work focuses on engineering feature sets for learning algorithms ranging from linear regression to more complex ensemble methods. While the Kaggle competition provides foundational game statistics, transforming these into effective predictors requires significant effort. The following subsection details the specific data types provided by the competition to establish this baseline. The final subsection we review existing literature with a primary focus on feature engineering. Finally, the paper details the implementation of distinct modeling techniques. This comparative analysis examines linear and logistic regression alongside additional statistical methods. These models are compared against one another and evaluated relative to the winning scores from prior competitions.

Data Available

The highly quantifiable nature of basketball ensures that statistical data is both abundant and accessible. While the Kaggle competition permits participants to incorporate any external datasets they acquire, the organizers themselves provide a comprehensive archive. This repository includes over 30 distinct tables of historical data spanning more than three decades. All of the tables are separated by gender, with the only exception being the public rankings table. This is due to the fact that they only compiled mens rankings.

The data provided by Kaggle is split into regular season and postseason files. For the purposes of this project, the tournament phase is referred to as the playoffs. This research focuses primarily on the detailed results tables for both the regular season and the playoffs while also utilizing the essential rankings and seeding tables. Although the broader archive extends further back, the detailed game statistics required for this analysis commence in the 2003 season. Each entry in these tables provides a comprehensive statistical breakdown for both the winning and losing teams in every recorded matchup.

Season	DayNum	WTeamID	WScore	LTeamID	LScore	WLoc	NumOT	WFGM	WFGA	WFGM3	WFGA3	WFTM	WFTA	WDR	WDR	WAsst	WTO	WSR	WBlk	WPF	LFGM	LFGA	LFGM3	LFGA3	LFTM	LFTA	LOR	LDR
2003	10	1104	68	1328	62	N	0	27	58	3	14	11	18	14	24	13	23	7	1	22	22	53	2	10	16	22	10	22
2003	10	1272	79	1393	63	N	0	26	62	8	20	10	19	15	28	16	13	4	4	18	24	67	6	24	9	20	20	25
2003	11	1296	73	1437	61	N	0	24	58	8	18	17	29	17	26	15	10	5	2	25	22	73	3	26	14	23	31	22
2003	11	1296	56	1457	50	N	0	18	38	3	9	17	31	6	19	11	12	14	2	18	18	49	6	22	8	15	17	20
2003	11	1400	77	1208	71	N	0	30	61	6	14	11	13	17	22	12	14	4	4	20	24	62	6	16	17	27	21	15
2003	11	1458	81	1186	55	H	0	26	57	6	12	23	27	12	24	12	9	9	3	18	20	46	3	11	12	17	6	22
2003	12	1161	89	1236	62	H	0	23	55	2	8	32	39	13	18	14	17	11	1	25	19	41	4	15	20	28	9	21
2003	12	1186	75	1457	61	N	0	28	62	4	14	15	21	13	35	19	19	7	2	21	20	59	4	17	17	23	8	25
2003	12	1194	71	1156	66	N	0	28	58	5	11	10	18	9	22	9	17	9	2	23	24	52	6	18	12	27	13	26
2003	12	1458	84	1296	56	H	0	32	67	5	17	15	19	14	22	11	6	12	0	13	23	52	3	14	7	12	9	23
2003	13	1166	106	1426	59	H	0	41	69	15	25	9	13	15	29	21	11	10	6	16	17	52	4	11	12	17	8	15
2003	13	1202	74	1106	73	N	0	29	51	7	13	9	11	6	21	18	15	7	1	5	29	63	10	22	5	5	13	16
2003	13	1237	66	1135	65	N	0	26	66	5	19	9	13	21	23	15	17	12	3	17	24	56	6	19	11	17	14	21
2003	13	1323	76	1125	48	H	0	25	56	10	23	16	23	8	35	18	13	14	19	13	18	64	8	24	4	8	14	26
2003	14	1125	83	1135	77	N	1	30	70	11	31	12	15	15	18	22	16	18	5	21	28	69	4	15	17	27	20	27
2003	14	1156	78	1236	71	N	0	27	46	10	18	14	24	8	25	18	15	2	6	18	23	50	7	20	18	24	6	17
2003	14	1161	81	1194	56	H	0	22	48	5	12	32	43	16	32	13	24	5	3	19	21	65	2	13	12	19	16	18
2003	14	1186	82	1202	57	H	0	33	61	8	19	8	14	9	23	21	17	14	5	18	23	55	2	11	9	14	13	23
2003	14	1183	73	1129	59	A	0	29	59	9	19	6	8	8	21	17	15	10	1	19	21	51	3	14	14	18	12	22
2003	14	1314	85	1336	55	H	0	32	60	5	15	16	23	12	34	18	14	8	4	12	21	71	3	23	10	11	15	20
2003	14	1323	89	1237	45	H	0	34	62	8	16	13	21	12	34	25	11	9	13	16	18	74	4	15	5	11	27	22
2003	14	1353	69	1162	36	H	0	23	57	4	19	10	15	15	18	10	12	14	6	18	12	39	3	8	9	13	11	23
2003	14	1390	61	1131	57	H	0	20	53	6	27	15	24	15	24	9	14	4	2	17	19	54	3	17	16	18	11	22
2003	14	1426	59	1106	47	N	0	25	53	1	5	8	12	15	32	14	16	6	4	7	20	60	6	18	1	2	8	15
2003	14	1462	87	1389	48	H	0	35	70	6	19	11	16	17	27	18	7	13	3	10	20	63	7	20	1	4	16	18
2003	15	1161	77	1156	63	H	0	27	58	5	10	18	27	18	26	15	12	5	4	21	23	51	3	13	14	20	7	19
2003	15	1194	93	1236	86	N	2	33	75	9	20	18	33	17	26	14	14	17	0	21	27	58	14	31	18	22	9	33
2003	15	1196	76	1256	55	H	0	28	48	4	13	16	24	8	23	18	18	13	5	19	19	49	1	5	16	22	12	17
2003	15	1242	81	1221	57	H	0	28	53	3	5	22	37	13	29	12	19	10	3	21	18	52	7	21	14	21	8	22
2003	15	1422	84	1447	65	H	0	33	53	3	8	15	20	7	36	12	22	5	9	19	26	70	3	25	10	16	14	15
2003	16	1314	71	1353	67	H	0	27	57	5	14	12	19	14	30	19	20	8	5	13	27	66	9	22	4	11	12	19
2003	16	1390	63	1462	62	H	0	23	57	6	15	11	21	17	20	12	10	8	1	19	18	45	6	20	20	28	11	17
2003	17	1196	99	1183	65	H	0	39	73	11	28	10	22	16	31	32	14	10	14	19	22	66	4	11	17	24	15	23
2003	17	1304	68	1147	45	N	0	27	58	6	18	8	13	10	28	13	17	5	5	26	10	50	2	16	23	31	16	24
2003	18	1104	82	1186	56	H	0	24	48	10	20	24	34	12	26	13	13	7	3	14	19	55	8	21	10	12	11	18

Figure 1. Regular Season Detailed Data

This table records key performance metrics for each game played. The schema utilizes the prefix W to denote statistics for the winning team and the prefix L for the losing team. The specific recorded variables include:

- Points Scored
- The number of field goals made and attempted (FGM-A).
- The number of 3-point field goals made and attempted (FGM-A3).

- The number of free throws made and attempted (FTM-A).
- The number of offensive and defensive rebounds secured (OR- DR).
- The number of assists (Ast).
- The number of turnovers (TO).
- The number of blocks (Blk).
- The number of fouls (PF).

While raw box scores provide the foundation of events that happened in a game, they often fail to capture nuances in, and out, of the game. To address this limitation, this research supplements the Kaggle datasets with advanced metrics from analyst Ken Pomeroy (KenPom). Pomeroy is accepted as the industry standard for college basketball analysis. Pomeroy's approach is to normalize the data by calculating efficiency rather than raw stats. These statistics are updated daily throughout the regular season and tournament. For the purpose of this study, a snapshot of the rankings was acquired from the day prior to the start of the 2025 tournament.

2026 Pomeroy College Basketball Ratings														
help														
02-03-04-05-06-07-08-09-10-11-12-13-14-15-16-17-18-19-20-21-22-23-24-25-26														
Data through games of Monday, January 19 (3689 games)														
Rk	Team	Conf	W-L	NetRtg	ORTg	DRtg	AdjT	Luck	Strength of Schedule			NCSOS		
									NetRtg	ORTg	DRtg			
1	Michigan	B10	16-1	+36.00	127.0	91.0	73.5	+0.22	144	+11.23	116.6	105.4	+11.45	14
2	Arizona	B12	18-0	+34.04	126.9	92.9	71.8	+0.33	122	+6.59	114.8	108.2	+2.69	115
3	Duke	ACC	17-1	+32.78	126.0	93.2	67.6	+0.78	43	+9.69	115.7	106.0	+6.04	51
4	Purdue	B10	17-1	+31.31	129.5	98.2	65.9	+0.87	35	+9.41	116.1	106.7	+7.23	36
5	Houston	B12	17-1	+31.14	124.1	93.0	63.6	+0.54	78	+5.44	113.5	108.1	+0.85	159
6	Gonzaga	WCC	19-1	+31.04	126.3	95.2	70.3	+0.96	23	+5.48	113.1	107.7	+7.66	33
7	Illinois	B10	15-3	+30.44	129.3	98.9	66.4	+0.04	180	+8.62	115.5	106.9	+3.56	93
8	Iowa St.	B12	16-2	+30.00	125.0	95.0	68.2	+0.33	121	+5.29	112.3	107.0	-2.15	246
9	Florida	SEC	13-5	+29.57	125.6	96.1	70.4	-0.90	343	+12.33	117.9	105.6	+6.97	41
10	Michigan St.	B10	16-2	+29.32	120.0	90.7	66.3	-0.32	256	+8.60	116.7	108.1	+2.69	114
11	Connecticut	BE	18-1	+28.74	121.9	93.2	65.5	+0.87	34	+9.44	114.6	105.2	+7.69	32
12	Vanderbilt	SEC	16-2	+28.13	127.1	98.9	71.2	+0.32	123	+7.96	115.1	107.1	+1.72	134
13	Nebraska	B10	18-0	+27.06	123.5	96.5	67.3	+0.92	29	+5.94	113.8	107.9	-4.60	296
14	Virginia	ACC	16-2	+26.84	124.7	97.9	65.8	+0.02	185	+4.52	113.2	108.7	-3.64	279
15	BYU	B12	16-2	+26.70	125.2	98.5	70.8	+0.40	105	+7.19	113.9	106.8	+3.71	86
16	Louisville	ACC	13-5	+26.05	126.9	100.9	70.3	-0.54	300	+7.76	113.2	105.4	+2.11	124
17	Alabama	SEC	13-5	+25.46	129.4	104.0	73.8	-0.04	198	+14.83	119.1	104.2	+12.50	9
18	Iowa	B10	13-5	+25.37	123.0	97.7	63.9	-0.70	321	+4.71	112.3	107.6	-7.12	337
19	Kansas	B12	13-5	+25.25	122.0	96.8	67.8	-0.16	232	+12.99	116.4	103.4	+10.51	17
20	St. John's	BE	13-5	+24.44	123.1	98.7	70.9	-0.64	312	+9.28	115.8	106.5	+7.09	40

Figure 2. The first 20 teams of the KenPom data table.

The primary metrics utilized from this source include:

- Net Rating (NetRtg).
- Offensive Rating (ORTg).
- Defensive Rating (DRtg).
- Adjusted Tempo (AdjT).

Net rating is the differential between Offensive and Defensive Rating.

Offensive Rating measures the points scored per 100 possessions. Similarly, defensive rating measure the points given up per 100 possessions. These ratings are adjusted by the quality of opponents a team has played, and the game location. Lastly, Adjusted tempo is the number of possessions per 40 minutes. We define a possession as the period where a team controls the ball, ending with a shot attempt, turnover, or score.

- Adjusted Offensive Efficiency (ADJOE).
- Adjusted Defensive Efficiency (ADJDE).
- Adjusted Tempo (ADJ T.).

RK	TEAM	CONF	G	REC	D-I AVG:		110.2		0.65		EFF. FGM %		TURNOVERS 17.2		REBOUNDS 30.6		FT RATE 56.6		3-PT % 51.7		3-PT % 51.8		SP RATE 35.7		3PT % 47.7		WAB
					ADJUE	ADJUE	BARTHING	ADJUE	ADJUE	ADJUE	ADJUE	ADJUE	ADJUE	ADJUE	ADJUE	ADJUE	ADJUE	ADJUE	ADJUE	ADJUE	ADJUE	ADJUE	ADJUE	ADJUE	ADJUE	ADJUE	
1	Michigan	B10	17	16-1	128.8	91.0	.8818		59.3	42.3	16.1	17.2	35.0	28.4	70	26	1	63.9	40.4	35.4	30.1	42.8	41.7	73.8	7	+5.6	
2	Houston	B12	18	17-1	125.4	92.7	.9701		52.1	45.7	13.2	23.7	38.8	30.9	25.0	38.1	52.8	45.7	34.2	30.5	42.1	43.3	63.6	4	+8		
3	Arizona	B12	18	18-0	125.8	93.6	.9676		57.0	45.8	16.2	18.2	40.3	25.0	43.1	29.1	57.9	44.3	38.5	32.3	27.4	36.1	71.8	3	+6.3		
4	Connecticut	BE	19	18-1	122.2	92.4	.9616		55.3	43.6	16.6	19.5	34.6	29.1	31.9	38.8	57.2	44.8	34.7	27.5	37.6	33.8	65.3	5	+6.4		
5	Purdue	B10	18	17-1	129.3	98.1	.9598		58.8	48.8	14.2	16.4	37.0	26.7	28.1	22.1	58.9	51.4	39.2	30.6	38.4	44.5	65.7	5	+5.7		
6	Illinois	B10	18	15-3	130.6	99.3	.9591		55.8	45.9	14.6	12.3	39.7	27.5	33.8	19.1	58.8	45.1	35.1	31.4	48.6	40.6	66.3	4	+4.0		
7	Vanderbilt	SEC	18	16-2	128.6	97.8	.9586		57.8	46.8	13.1	18.7	38.0	27.5	38.1	43.0	59.9	49.1	38.8	28.8	44.3	40.3	70.9	7	+4.3		
8	Virginia	ACC	18	16-2	126.5	97.0	.9546		56.0	43.5	15.8	16.4	40.6	30.4	36.4	35.2	56.3	31.6	37.1	28.9	40.5	37.2	65.6	3	+3.8		
9	Florida	SEC	18	13-5	125.7	96.6	.9542		52.4	47.1	17.0	17.2	43.5	23.2	40.8	36.0	59.0	45.5	26.4	33.6	40.4	33.5	70.0	7	+2.6		
10	Zongaza	WCC	20	19-1	124.8	96.0	.9536		57.7	45.6	14.5	20.5	37.3	24.5	30.6	32.0	58.8	46.6	36.8	29.8	29.9	44.7	70.3	5	+5		

Figure 3. The first 10 teams of the Barttorvik table

To complement the efficiency metrics provided by Pomeroy and Torvik, this research also incorporates the comprehensive ranking systems aggregated by Kenneth Massey. While the previously discussed datasets focus on raw predictive efficiency, the Massey Ordinals provide a consolidated view of team hierarchy. This dataset aggregates rankings from over 40 distinct

mathematical systems, pollsters, and algorithms. These rankings typically present data as an ordinal integer where the top team is assigned the value of 1. Incorporating this data allows the model to leverage a "wisdom of the crowd" approach by utilizing the consensus of the broader analytical community.

TEAM	SORT	DP	EBP	EMK	ENG	ESR	FAS	HAS	INC	JNG	KPK	LOG	MAS	MB	MGS
Arizona	>>	4	5	2	2	1	1	4	2	1	3	2	1	1	2
Michigan	>>	1	2	1	1	6	3	1	1	4	1	1	6	6	4
Duke	>>	2	1	3	7	4	4	2	3	2	2	3	2	5	5
Purdue	>>	7	6	5	3	5	6	3	5	6	4	5	4	4	3
Gonzaga	>>	3	3	9	9	7	7	7	6	9	5	4	5	7	1
Connecticut	>>	13	11	6	5	3	5	15	7	5	6	7	7	3	6
Houston	>>	5	4	4	6	8	9	8	4	7	8	12	3	8	9

Figure 4. A sample of the Massey Ordinals table showing distinct ranking systems.

One significant challenge in aggregating these distinct sources is the inconsistency in team nomenclature. For instance, a university might be listed as "St. Mary's" in one dataset and "Saint Mary's CA" in another. To resolve this discrepancy, all external data was strictly mapped to the unique TeamID integer provided by the Kaggle competition files. This standardization ensures that the advanced efficiency metrics are accurately merged with the box score data for every specific matchup.

LITERATURE REVIEW

Past Strategies

The pursuit of a perfect bracket has generated a substantial body of research in predictive modeling over the last decade. While specific algorithms vary, the most successful strategies in the Kaggle Machine Learning Mania competition consistently rely on one of two distinct methodological frameworks.

The primary objective for this study is to minimize the Brier Score. We define the Brier Score (BS) as follows:

$$(1) \quad BS = \frac{1}{N} \sum_{t=1}^N (f_t - o_t)^2$$

where N is the number of events, f_t is the predicted probability, and o_t is the actual outcome. For this research, a 0 is considered a loss, and a 1 is a win. A Brier Score of 0 would reflect perfect accuracy, while a Brier Score of 1 reflects perfect inaccuracy. To provide a baseline, a strategy where we predict every team has a 50% chance of winning would result in a Brier Score of 0.25. (thinking of putting another example here)

The first framework relies on fundamental analysis where the objective is to construct a ground-truth probability independent of public sentiment. A common implementation of this approach involves utilizing domain-specific heuristics such as the elo rating system. Originally designed for chess, elo provides a dynamic measure of team strength that updates after every game. It offers a robust baseline that accounts for strength of schedule more

effectively than raw win-loss records [10]. Many strategies also incorporate expert features. These include the Nate Silver, FiveThirtyEight, and KenPom ratings which act as high-signal inputs that aggregate player health, travel distance, and roster depth into a single numeric value. By treating these expert outputs as features rather than final predictions, modelers can leverage human domain expertise while correcting for known expert biases.

The second dominant framework relies on the Efficient Market Hypothesis [8]. This approach posits that global betting markets, specifically closing lines from major bookmakers, offer a more accurate aggregate predictor of game outcomes than individual statistical models. The efficacy of this strategy is well-documented in recent Kaggle competitions where five of the top eight performing models utilized market-derived features as their primary feature. These strategies do not attempt to beat the market. Instead, they focus on mathematically disentangling the true win probability from the bookmaker’s overround and inherent biases.

To achieve this, recent state-of-the-art implementations utilize specific conversion algorithms to de-noise betting data. The standard tool in this domain is the *Goto_conversion*. Competitors employ these functions to address the favorite-longshot bias. This is the tendency for markets to overvalue underdogs and undervalue favorites. By applying these corrections, modelers can transform raw Vegas odds into highly calibrated probabilities that minimize Brier Score more effectively than fundamental analysis alone. This approach effectively leverages the crowd wisdom of the market while stripping away the pricing inefficiencies introduced by public betting behavior. For more information on how this strategy works, we refer the reader to [9].

The applicability of these methodological frameworks, however, is heavily constrained by the structural discontinuity introduced by the 2021 Name, Image, and Likeness (NIL) legislation. Statistical analysis of the post-NIL landscape reveals a significant deviation from historical norms, characterized by a rapid consolidation of talent where the number of programs with elite win percentages ($> 75\%$) has nearly tripled [6]. This phenomenon implies that feature weights derived from pre-2021 eras are likely subject to significant concept drift. Consequently, this literature review restricts its focus to winning strategies from recent Kaggle competitions (2022-present), as methodologies developed before the NIL era may not reflect the approaches best suited to the heightened stratification of team strength driven by current transfer portal dynamics.

Modeling Techniques

Recent tournament prediction methodologies employ several established modeling and computational techniques. This section provides formal definitions for the five core approaches used in this study: Logistic Regression, Linear Regression, Monte Carlo Simulation, Random Forest, and Neural Networks.

Logistic Regression is a binary classification model that estimates the probability of an outcome by applying the logistic (also known as sigmoid) function to a linear combination of input features. The model is defined as:

$$(2) \quad P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n)}}$$

where Y is the binary outcome, $X = (x_1, x_2, \dots, x_n)$ represents the feature vector, and $\beta = (\beta_0, \beta_1, \dots, \beta_n)$ are the model coefficients estimated through the maximum likelihood estimation. This function ensures the predicted probabilities are bounded between $[0, 1]$. Logistic regression assumes a linear relationship between the input features and the log-odds of the outcome. This assumption can be restrictive when the true relationships are nonlinear. To address this limitation, common approaches are feature engineering with polynomial or interaction terms (e.g., $x^2, x_i \cdot x_j$) to capture nonlinear patterns.

Linear Regression is a supervised learning method that models the relationship between a dependent variable and one or more independent variables by fitting a linear equation to the observed data. Linear Regression is defined as:

$$(3) \quad \hat{Y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

where $\hat{Y}$ is the predicted continuous outcome, $X = (x_1, x_2, \dots, x_n)$ represents the vector of features, and $\beta = (\beta_0, \beta_1, \dots, \beta_n)$ are the model coefficients estimated through ordinary least squares (OLS) minimization. For more information on OLS we refer the reader to [11]. Unlike logistic regression, linear regression produces unbounded predictions and does not inherently output probabilities. In sports prediction contexts, linear regression is commonly used to estimate point totals, scoring margins, or other continuous performance metrics that can subsequently be transformed into win probabilities through additional techniques.

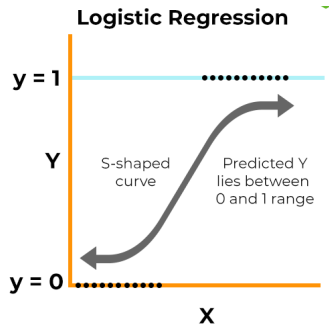


Figure 5. An Example of a Logistic Regression Equation [12].

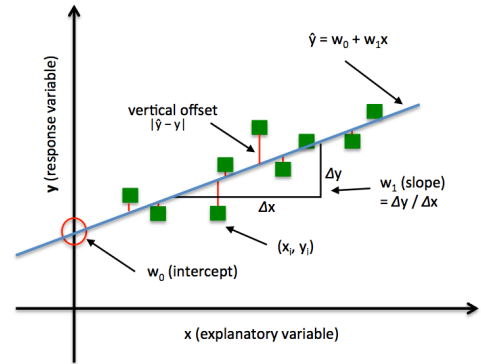


Figure 6. An Example of a Linear Regression Equation [13].

Monte Carlo Simulation is a computational technique that uses repeated random sampling to estimate probabilities for systems with inherent uncertainty. The method generates N independent samples $x_1, x_2, \dots, x_N$

from a probability distribution $f(x)$, then estimates probabilities by computation the proportion of samples satisfying a certain condition. The simulation is defined as:

$$(4) \quad \hat{P} = \frac{1}{N} \sum_{i=1}^N 1(x_i \in A)$$

where $1(\cdot)$ is the indicator function that equals 1 when the condition is satisfied and 0 otherwise. A represents the event of interest, and $\hat{P}$ is the Monte Carlo estimate of the true probability $P(A)$. By the Law of Large Numbers [14], $\hat{P} \rightarrow P(A)$ as $N \rightarrow \infty$. In sports prediction contexts, Monte Carlo methods are frequently employed to simulate game outcomes by sampling from estimated score distributions, enabling the estimation of win probabilities through empirical frequency counts across simulated matchups.

XGBoost (Extreme Gradient Boosting), considered a "black-box" approach, is an ensemble learning method that constructs a sequence of decision trees where each successive tree is trained to correct the residual errors of the previous trees. Unlike Random Forest, which builds trees independently, XGBoost employs gradient boosting to iteratively minimize a loss function through additive modeling [5]. The model prediction is defined as the sum of predictions from all trees:

$$(5) \quad \hat{y}_i = \sum_{k=1}^K f_k(x_i).$$

Here, $f_k(x_i)$ is the prediction of the k^{th} tree for the i^{th} data point, $\hat{y}_i$ is the final predicted value for the i^{th} data point, and K is the number of trees.

Each tree is trained to minimize the objective function

$$(6) \quad \text{obj}(\theta) = \sum_i^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k)$$

where l is the loss function which simply computes the difference between the true and predicted, and Ω is the regularization term, discouraging overly complex trees. The optimization process then proceeds via gradient descent on the residuals. For more information regarding gradient descent we refer the reader to [4]. XGBoost is widely adopted in competitive machine learning for its performance on structured data and ability to capture complex interactions and relationships between features.

$$(7) \quad h^{(\ell)} = \sigma(W^{(\ell)}h^{(\ell-1)} + b^{(\ell)})$$

if i do neural nets i will keep going with this.

Probability Prediction Models

This section details the development and evaluation of models designed to predict a continuous win probability $P \in (0, 1)$ for any given matchup. While the tournament results are ultimately binary, the Brier Score evaluation metric heavily penalizes overconfidence in incorrect predictions and rewards well-calibrated probabilities. Consequently, this research moves beyond simple classification to focus on models that quantify the uncertainty inherent in the March Madness environment. By establishing a framework where a value $P > 0.5$ denotes a predicted victory for Team A, a baseline is created for performance that accounts for the high variance of single-elimination play. This probabilistic approach serves as the engine of the project and bridges the gap between theoretical literature and the empirical application of machine learning.

To expand further on these concepts, the methodology begins with a rigorous data processing phase where historical box scores are transformed into a symmetric training set. This ensures that the models learn to evaluate the relative strength between two competitors rather than relying on arbitrary formatting artifacts. This section details the complete machine learning pipeline implemented for the 2026 tournament prediction. We first examine the data cleaning procedures required to tailor the feature sets to the specific statistical assumptions of each algorithm. Following this, we explore the variable selection process used to identify high signal metrics such as KenPom and Massey data. Finally, we implement and compare two distinct modeling architectures, specifically Logistic Regression and XGBoost, to determine

which approach best minimizes the Brier Score in a high variance environment.

Logistic Regression

Variable Selection

To accurately model basketball outcomes, one must first deconstruct the sport into its atomic statistical events. A standard game is composed of a finite sequence of possessions. Within these possessions, specific discrete actions determine the flow of the contest. The primary events recorded in the raw dataset include Field Goals, Free Throws, Rebounds, Turnovers, and Personal Fouls. A Field Goal (*FGM*) represents the primary method of scoring and is worth either two or three points. A Free Throw (*FTM*) is an unopposed attempt worth one point awarded after a foul. A Rebound is recorded when a player retrieves the ball after a missed shot; this is further categorized into Offensive Rebounds (*OR*), which grant the shooting team a second opportunity, and Defensive Rebounds (*DR*), which terminate the opponent's possession. Finally, a Turnover (*TO*) is a loss of possession due to a violation or steal, resulting in zero points.

Analyzing these events using raw totals is statistically flawed due to the confounding variable of pace. A team that plays at a high tempo may record high totals for points and rebounds simply because they generate more possessions, not because they are efficient. To resolve this, this model utilizes the "Four Factors of Basketball Success." Developed by Dean Oliver, this framework isolates the four specific components that correlate most strongly with winning: shooting efficiency, turnover avoidance, rebounding dominance, and free throw frequency [16].

Crucial to these efficiency calculations is the estimation of total Possessions ($Poss$). A possession is defined as the period during which a team controls the ball, ending with a shot attempt, a turnover, or a trip to the free-throw line. Since possessions are not explicitly tracked in standard box scores, they must be estimated using the following formula:

$$(8) \quad Poss = (FGA - OR) + TO + (0.44 \cdot FTA).$$

This equation aggregates the three ways a possession can end: a field goal attempt (FGA), a turnover (TO), or free throws (FTA). The term $(FGA - OR)$ accounts for missed shots that are rebounded by the offense; because the team retains the ball, the possession continues, and thus the initial attempt is subtracted to avoid double-counting. The coefficient 0.44 applied to Free Throw Attempts estimates the proportion of trips to the line that actually terminate a possession (e.g., the second of two shots), accounting for and-one (fouled while successfully scoring) situations and technical fouls. This derived metric serves as the denominator for rate statistics, allowing for a direct comparison between teams of differing tempos.

The first factor is Effective Field Goal Percentage ($eFG\%$). This metric refines standard field goal percentage by accounting for the fact that a three-point shot is worth 50% more than a two-point shot. It is calculated as:

$$(9) \quad eFG\% = \frac{FGM + 0.5 \cdot FGM3}{FGA}$$

By including the 0.5 weighting for three-pointers ($FGM3$), this variable measures a team's ability to maximize points per attempt regardless of their shot selection strategy. The second factor is Turnover Percentage ($TOV\%$). Unlike raw turnover counts, this metric normalizes errors against the total number of possessions calculated above.

$$(10) \quad TOV\% = \frac{TO}{Poss}$$

This ratio identifies teams that value the basketball. A lower percentage indicates a disciplined offense that rarely forfeits scoring opportunities. The third factor is Offensive Rebound Percentage ($ORB\%$). This metric measures a team's physical dominance relative to their available opportunities. Using raw offensive rebound totals is misleading because a team that misses many shots will naturally have more chances to rebound. This metric corrects for that bias by comparing offensive rebounds (OR) to the total number of rebounds available at that specific rim:

$$(11) \quad ORB\% = \frac{OR}{OR + DR_{opp}}$$

Here, DR_{opp} represents the defensive rebounds secured by the opponent. A higher percentage indicates a team that frequently generates "second chance" points. Finally, the model incorporates Win Percentage ($Win\%$) as a fifth feature to capture intangible qualities such as coaching and clutch performance. This is calculated as the mean of the binary game results (1 for a win, 0 for a loss) over the course of the season. Crucially, the logistic

regression algorithm does not process these values in isolation. Instead, the dataset is transformed to calculate the differential between the two competing teams for every metric (e.g., $eFG\%_{diff} = eFG\%_{TeamA} - eFG\%_{TeamB}$). This centers the analysis on the relative mismatch between the opponents rather than their absolute statistics.

Data Cleaning

To construct a robust predictive pipeline, the data architecture must establish a strict boundary between feature generation and model training. The independent variables, specifically the Four Factors and Win Percentage, are calculated exclusively utilizing historical regular-season detailed results. Conversely, the logistic regression algorithm is trained entirely on historical NCAA tournament outcomes. By isolating regular-season efficiency metrics as the predictors and tournament game results as the target variable, the framework ensures that the model evaluates how a team's sustained performance over a thirty-game span translates to the high-variance, single-elimination environment of postseason play. This distinction is critical, as regular-season data provides a sufficiently large sample size to stabilize efficiency metrics, while tournament data provides the specific contextual outcomes the model ultimately aims to predict.

Furthermore, to prevent data leakage and simulate the exact informational constraints of a real-world forecaster, the logistic regression model is trained using a strictly defined rolling time window. To evaluate historical accuracy and construct valid out-of-sample predictions, the framework is systematically back tested across consecutive tournament years.

For example, to generate probability forecasts for the 2025 NCAA Tournament, the model is trained exclusively on the tournament matchups from the preceding 2023 and 2024 seasons. The decision to restrict the training data to a brief temporal window is grounded in the structural realities of collegiate athletics. NCAA basketball is characterized by high roster turnover, as student-athletes possess limited eligibility and typically serve as primary contributors for only three to four seasons. Consequently, a university program undergoes a complete personnel and strategic transformation within a short cycle. Training the algorithm on data exceeding this window would introduce noise from past iterations of a team that share no roster overlap with the current squad. During this localized training phase, the algorithm maps the recent regular-season metrics to their respective tournament results, learning the optimal coefficient weights without accessing any data from the target prediction year. This rolling training window shifts iteratively for historical benchmark predictions, ensuring that all evaluations remain mathematically rigorous and entirely devoid of look-ahead bias.

The primary obstacle in processing the historical game records is the fundamental structure of the Kaggle dataset. The raw regular-season and tournament data files utilize a deterministic outcome schema. Within this structure, every game observation is explicitly bifurcated into winning and losing statistics. Metrics belonging to the victorious program are assigned column headers prefixed with a 'W', while the defeated opponent's metrics are prefixed with an 'L'.

If a classification algorithm is trained directly on this format, it inevitably suffers from severe target leakage. The logistic regression algorithm

would trivially learn to associate the column position with the outcome, mathematically recognizing that the entity occupying the 'W' feature space possesses a 100 percent win probability. This renders the model entirely incapable of out-of-sample prediction, as the ultimate objective is to forecast matchups between two neutral entities before the outcome is known. The following output illustrates this problematic data architecture, demonstrating how the game result is inherently encoded into the feature presentation itself.

```
head(mens_results %>%
      select(Season, WTeamID, WScore,
             LTeamID, LScore, WFGM,
             LFGM, WTO, LTO), 4)
```

##	Season	WTeamID	WScore	LTeamID	LScore	WFGM	LFGM	WTO	LTO
##	<int>	<int>	<int>	<int>	<int>	<int>	<int>	<int>	<int>
## 1:	2003	1104	68	1328	62	27	22	23	18
## 2:	2003	1272	70	1393	63	26	24	13	12
## 3:	2003	1266	73	1437	61	24	22	10	12
## 4:	2003	1296	56	1457	50	18	18	12	19

Table 1. Snapshot of raw Kaggle regular-season data demonstrating the 'W' and 'L' column structure.

To resolve the target leakage inherent in the raw dataset, the first phase of data processing transforms the game-level observations into season-level team averages. This transformation is executed via the `calculate_four_factors` function. Initially, the function estimates the total possessions for both the winning and losing teams using the standard formula. Subsequently, it computes the Four Factors (Effective Field Goal Percentage, Turnover Percentage, Offensive Rebound Percentage, and Free Throw Rate) for each team in every game.

Rather than maintaining the winner and loser statistics in a single row, the function separates the victors and the defeated into two distinct dataframes. A binary **Win** variable is appended, assigning a value of 1 to the winning dataframe and 0 to the losing dataframe. These two structures are then vertically concatenated using a row-bind operation. Finally, the data is grouped by **Season** and **TeamID** to calculate the mean of all numeric variables. This yields a single, stable efficiency profile and an aggregate regular season win percentage for every program in a given year.

Model Implementation

Following the systematic cleaning and neutralization of the dataset, the analysis proceeds to the implementation of the predictive framework. The primary objective is to construct a logistic regression pipeline capable of generating calibrated win probabilities for every possible matchup in the 2026 NCAA Tournament. This method is chosen specifically for its ability to model the probability of a binary outcome, in this case whether the lower-ID team (Team 1) wins or loses, based on a linear combination of statistical differentials. The models utilize a logit link function to map the relationship between the independent variables and the probability of a victory. The formal specification for each model iteration is defined by the following equation:

(12)

$$\begin{aligned} \text{logit}(P) = \ln \left(\frac{P}{1-P} \right) = & \beta_0 + \beta_1(\text{eFG_diff}) + \beta_2(\text{TOV_diff}) \\ & + \beta_3(\text{ORB_diff}) + \beta_4(\text{FTR_diff}) + \beta_5(\text{Win_diff}) \end{aligned}$$

Where P represents the probability of a Team 1 victory. The coefficients (β) represent the change in log-odds of winning for every one unit increase in the respective statistical differential.

The model is estimated using maximum likelihood estimation, which identifies the coefficient vector β that maximizes the probability of observing the recorded tournament outcomes given the input differentials. Because the response variable is binary, the log-likelihood function takes the form:

$$(13) \quad \sum_{i=1}^n [y_i \log p(x_i) + (1 - y_i) \log(1 - p(x_i))]$$

where $y_i \in \{0, 1\}$ is the observed outcome for the i -th matchup and $p(x_i)$ is the model's predicted probability [17] of a Team 1 victory.

Maximizing this function is equivalent to minimizing the binary cross-entropy loss. The resulting coefficients reflect both the direction and magnitude of each differential's contribution to the probability of winning. A positive coefficient on **eFG_diff**, for instance, indicates that a larger shooting efficiency advantage is associated with a higher probability of victory, which is consistent with what basketball theory would predict.

For the 2026 deployment, separate models are trained for the men’s and women’s divisions. The choice to train independent models for each division rather than a single combined model with a gender indicator reflects a fundamental assumption that the Four Factors carry structurally different predictive weight in the men’s and women’s game. Differences in pace, physicality, and three-point volume between the two divisions suggest that the relationship between shooting efficiency and winning probability may not be uniform across genders. Training separate models allows each set of coefficients to reflect the unique dynamics of its respective division rather than forcing a shared parameter estimate that could mask meaningful differences.

Nonetheless, Each model is trained exclusively on tournament outcomes from the 2024 and 2025 seasons, with the corresponding regular-season Four Factors serving as the independent variables. This two-year window was selected to reflect the current competitive landscape of college basketball while respecting the structural roster turnover discussed previously. The Kaggle competition provides a submission file containing every possible 2026 tournament matchup, numbering in the thousands of pairings across both divisions. The Four Factors for each team are computed from their 2025 regular-season results and differenced for every candidate pairing. The fitted models then generate a probability $\hat{P} \in (0, 1)$ for each of these matchups via the inverse logit transformation, producing the final submission.

An important practical consideration in this implementation is the relatively small sample size introduced by the two-year training window. A single NCAA tournament produces only 63 games for the men’s bracket and

63 for the women’s, yielding a combined training pool of roughly 250 observations per model iteration. While this is sufficient for a low-dimensional model such as logistic regression, it places meaningful constraints on the complexity of the architecture that can be reliably estimated. Overfitting becomes a nontrivial risk if additional interaction terms or polynomial features are introduced without sufficient regularization. For this reason, the model is deliberately kept simple, relying on five theoretically motivated differentials rather than a broader set of engineered features. This simplicity is consistent with the principle of Occam’s Razor in model selection [18], where a simpler model that generalizes well out-of-sample is preferred over a complex one that fits the training data too closely.

Prior to submitting predictions for the 2026 tournament, the pipeline is validated through a series of backtests spanning the 2023, 2024, and 2025 competitions. These historical evaluations serve a specific purpose: to confirm that the rolling two-year training window generalizes to unseen tournament data before any live predictions are committed. For the 2023 backtest, the model is trained on tournament outcomes from 2021 and 2022. For the 2024 backtest, the training window shifts forward to include 2022 and 2023. Finally, for the 2025 backtest, the model is trained on 2023 and 2024 tournament data. In each case, the Four Factors are derived exclusively from the regular-season data preceding the target year, and no tournament results from the prediction year are ever introduced into the training process.

The resulting Brier Scores from these backtests are compared against the winning submission scores published by Kaggle for the corresponding competition years. This comparison provides a meaningful external

benchmark, situating the logistic regression framework within the broader landscape of competitor strategies. Table 2 summarizes the Brier Scores produced by the model across all three backtest years alongside the winning scores for their respective competitions.

Table 2. Logistic Regression Backtest Results vs. Kaggle Benchmarks

Tournament Year	Model Brier Score	Winning Brier Score
2023	0.21653	0.17371
2024	0.21064	0.05313
2025	0.18401	0.10411

It is important to note that the 2024 winning Brier Score of 0.05313 is not directly comparable to the scores recorded in 2023 and 2025. In that competition year, Kaggle introduced a significant structural change to the submission format. Rather than submitting a single set of game-by-game win probabilities, competitors were permitted to submit a portfolio of up to one hundred thousand distinct bracket entries. The final score was then computed by averaging the Brier Score across rounds rather than across individual games, placing substantially greater weight on later-round matchups where fewer games are played. This round-weighted averaging mechanism naturally compresses scores toward lower values relative to the standard per-game formulation. Consequently, the 0.05313 figure reflects a fundamentally different evaluation environment and should be interpreted as a benchmark specific to the 2024 format rather than as evidence of superior predictive accuracy over other competition years.

Setting aside the 2024 anomaly, the remaining backtests reveal a consistent trend of improvement. The model's Brier Score declined from

0.21653 in 2023 to 0.18401 in 2025, reflecting incremental gains in calibration as the pipeline was refined. Crucially, the 2025 backtest represents the most direct analog to the 2026 submission, as it uses the identical training structure and feature set that will be deployed. While the model trails the top competitors in each year, it remains well above the naive 0.25 baseline and demonstrates stable generalization across multiple tournament cycles. These results provided sufficient confidence in the pipeline to justify its deployment for the 2026 Kaggle submission.

Model Analysis

With the models estimated on the 2023 and 2024 tournament data, the resulting coefficients can be examined to evaluate which of the Four Factors carry meaningful predictive weight in each division. Because separate models were trained for the men’s and women’s tournaments, the coefficient structures can be analyzed independently and then compared to assess whether the Four Factors operate differently across divisions. Tables 3 and 4 present the estimated coefficients, p-values, and significance levels for the men’s and women’s models respectively.

Table 3. Logistic Regression Coefficient Estimates (Men’s 2025 Model)

Variable	Estimate	p-value	Sig.
Intercept	0.4352	0.0348	*
eFG Diff	7.7952	0.2244	
TOV Pct Diff	-20.5106	0.0901	.
ORB Pct Diff	12.3634	0.0027	**
FTR Diff	-1.4307	0.7527	
Win Diff	1.3253	0.4365	

Table 4. Logistic Regression Coefficient Estimates (Women’s 2025 Model)

Variable	Estimate	p-value	Sig.
Intercept	-0.0888	0.6657	
eFG Diff	17.4458	0.0016	**
TOV Pct Diff	-19.8212	0.0237	*
ORB Pct Diff	12.4308	0.0004	***
FTR Diff	3.9742	0.4420	
Win Diff	-1.8468	0.4172	

In the men’s model, the only variable to reach conventional statistical significance is Offensive Rebound Percentage, with an estimated coefficient of 12.3634 and a p-value of 0.003. The positive sign is theoretically consistent with Dean Oliver’s framework [16], as a team that dominates the offensive glass generates additional possessions while simultaneously denying the opponent easy transition opportunities. Turnover Percentage approaches significance at the 0.10 level with an estimate of -20.5106, and the negative direction is again theoretically correct, as a team that surrenders possessions more frequently than its opponent is expected to win less often. Effective Field Goal Percentage, Free Throw Rate, and Win Percentage all fail to reach significance in the men’s model, suggesting that within the narrow two-year training window, shooting efficiency differentials and win records carry limited additional explanatory power once rebounding and turnover control are accounted for. The intercept of 0.4352 is statistically significant at the 0.05 level, implying a slight structural tendency for the lower-ID team to win in the men’s tournament even after controlling for all efficiency differentials. This is likely an artifact of the team ID assignment convention rather than a genuine basketball phenomenon.

The women's model tells a different story. Three variables reach statistical significance: Offensive Rebound Percentage (12.4308, $p < 0.001$), Effective Field Goal Percentage (17.4458, $p = 0.002$), and Turnover Percentage (-19.8212, $p = 0.024$). The consistency of Offensive Rebound Percentage across both models is the single most robust finding of this analysis, as the near identical magnitude of the estimates (12.36 in the men's model and 12.43 in the women's model) across two independently trained models strongly suggests that second chance scoring dominance is a universal determinant of tournament success regardless of division. The women's intercept is effectively zero and insignificant, indicating no structural bias toward either team ordering in that division.

The most substantive structural difference between the two models is the role of Effective Field Goal Percentage. In the women's model, it is the second most significant predictor with a coefficient nearly twice the magnitude of the men's estimate. In the men's model, the same variable is statistically indistinguishable from zero. This divergence suggests that shooting efficiency differentials are a more decisive factor in women's tournament outcomes, where a meaningful gap in $eFG\%$ between two opponents translates more reliably into a victory. In the men's tournament, the compressed talent distribution and faster pace of play may reduce the marginal impact of a given efficiency advantage, as superior athleticism and shot creation ability can compensate for efficiency deficits in ways that are not captured by the Four Factors alone. This finding provides direct empirical support for the decision to train the two models independently rather than pooling them under a shared parameter structure. Had a combined model been used, the

women's strong shooting efficiency signal would have been diluted by the men's null result, and the men's rebound signal would have been the only surviving predictor across both divisions.

Free Throw Rate fails to reach significance in either model, which is consistent with the nature of the statistic itself. Unlike shooting efficiency or turnover rate, free throw volume tends to be relatively stable across teams at the tournament level, as elite programs generally reach the line at comparable rates. When the differential between two opponents is consistently narrow, the variable carries little discriminatory power and the model appropriately assigns it a negligible coefficient. Win Percentage similarly contributes no significant explanatory power in either model once the efficiency differentials are controlled for. This suggests that the Four Factors effectively subsume the information contained in raw win-loss records; a team's season-long efficiency profile captures the underlying quality of the program more precisely than its aggregate record, which can be inflated or deflated by schedule strength and margin of victory.

Results

The performance of the 2025 logistic regression models is evaluated along two dimensions: classification accuracy via the confusion matrices and probability calibration via the Brier Score. Figure 7 presents the confusion matrices for the men's, women's, and combined models.

The women's model achieves the strongest classification performance at 77.6% accuracy, followed by the combined model at 73.9% and the men's model at 70.1%. This ordering is consistent with the coefficient analysis in the

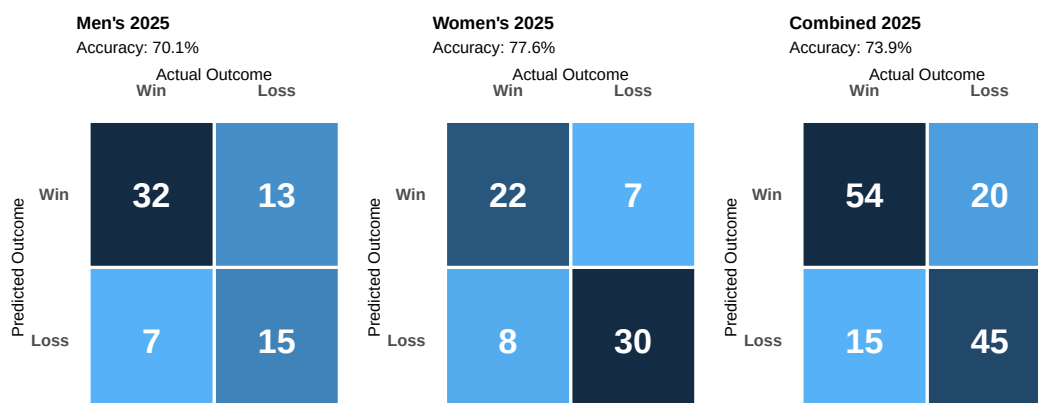


Figure 7. Confusion Matrices for the 2025 Logistic Regression Models

prior section. The women's model benefits from three statistically significant predictors, Offensive Rebound Percentage, Effective Field Goal Percentage, and Turnover Percentage, providing a richer signal for binary classification. The men's model, where only Offensive Rebound Percentage reaches conventional significance, has a narrower basis for discrimination and consequently misclassifies a greater proportion of games. All three accuracy figures comfortably exceed the 50% baseline that would be expected from random prediction, indicating that the Four Factors carry genuine discriminatory power in both divisions.

However, classification accuracy alone is an incomplete measure of model quality in this context. The Brier Score evaluation framework penalizes not just incorrect predictions but overconfident ones, meaning a model that correctly identifies the winner but assigns a probability of 0.95 rather than 0.65 is still penalized heavily when upsets occur. The combined 2025 Brier Score of 0.18401 reflects this tension. While the model demonstrates solid directional accuracy, the gap between its Brier Score and the competition

winning score of 0.10411 suggests that the predicted probabilities are not as well calibrated as those produced by the top competitors. This is an expected limitation of a parsimonious four factor framework. Models that incorporate additional signals such as KenPom efficiency ratings, seed differentials, or market-derived probabilities are better positioned to assign more conservative probabilities to matchups involving significant uncertainty, which is precisely where the Brier Score penalizes most heavily.

Despite this gap, the 2025 results validate the pipeline as a functional and reproducible baseline. The model generalizes consistently across both divisions, produces accuracy well above chance, and maintains a Brier Score substantially below the naive 0.25 baseline. These characteristics provide a foundation from which the more sophisticated modeling techniques discussed in the following sections aim to improve.

Boosting

While logistic regression provides an interpretable and theoretically grounded baseline, its core assumption of a linear relationship between the input differentials and the log-odds of winning imposes meaningful constraints on its predictive ceiling. In reality, the relationship between team efficiency metrics and tournament success may be nonlinear and subject to complex interactions. For instance, a team with a dominant offensive rebounding advantage may only convert that advantage into wins when paired with a sufficient shooting efficiency threshold, a conditional relationship that a linear model cannot capture without explicit feature engineering. To address this limitation, the analysis extends the modeling framework to include XGBoost.

XGBoost is a natural candidate for this problem for several reasons beyond its flexibility. Unlike parametric models, it does not require assumptions about the distributional form of the predictors or their relationship to the outcome. It is also robust to the scale differences between features, meaning the large coefficient magnitudes observed for Turnover Percentage and Offensive Rebound Percentage in the logistic regression output do not require normalization before fitting. Furthermore, XGBoost includes built-in regularization parameters that directly penalize model complexity, which is particularly valuable given the small training sample introduced by the two-year rolling window. The following subsections detail the variable selection, data cleaning, and implementation decisions made for the boosting framework, mirroring the structure of the logistic regression analysis to facilitate a direct comparison between the two approaches.

Variable Selection

A key motivation for the breadth of this feature set is that XGBoost’s internal regularization framework absorbs the cost of including additional variables in a way that ordinary regression cannot. In logistic regression, adding correlated or weakly predictive features inflates standard errors, destabilizes coefficient estimates, and risks overfitting, which is why the Four Factors model was deliberately restricted to five predictors. XGBoost does not share this vulnerability. At each boosting round the algorithm selects only the features and split points that produce the greatest reduction in the loss function, effectively ignoring features that contribute no signal. Features that are redundant or noisy are simply passed over during tree construction rather than distorting the model as they would in a linear framework. This property is reinforced by the `colsample_bytree` hyperparameter, which randomly samples a subset of features at each tree, further reducing the influence of any single uninformative predictor. As a result, the XGBoost feature set can be expanded aggressively without the same penalty that would apply in a parametric setting, and the cost of including a feature that turns out to be irrelevant is negligible compared to the cost of omitting one that carries genuine signal.

The feature set for the XGBoost model is organized into three tiers of increasing complexity: seeding features, simple season statistics, and strength of schedule adjusted box score profiles. Together these tiers provide the model with information about committee evaluation, raw performance, and opponent-adjusted efficiency. The first tier consists of tournament seeding. Each team’s seed number is included individually as `T1_seed` and `T2_seed`,

along with their difference **Seed_diff**, defined as the opponent's seed minus the team's seed such that a positive value indicates a favorable matchup. Including all three seed features rather than the differential alone allows the model to learn nonlinear effects of absolute seeding. A seed differential of one means something very different when it separates a 1 seed from a 2 seed than when it separates an 8 seed from a 9 seed. The lower seeded teams in the tournament are generally far more dominant over their opponents than the differential alone would suggest, and a model that only sees the difference cannot distinguish between these cases. Figure 8 confirms this, showing that average point differential declines monotonically as seed number increases across both divisions. Figure ?? further shows that as the seed differential between two teams widens, the better seeded team wins by larger margins on average, validating seed differential as a strong predictive signal in its own right.

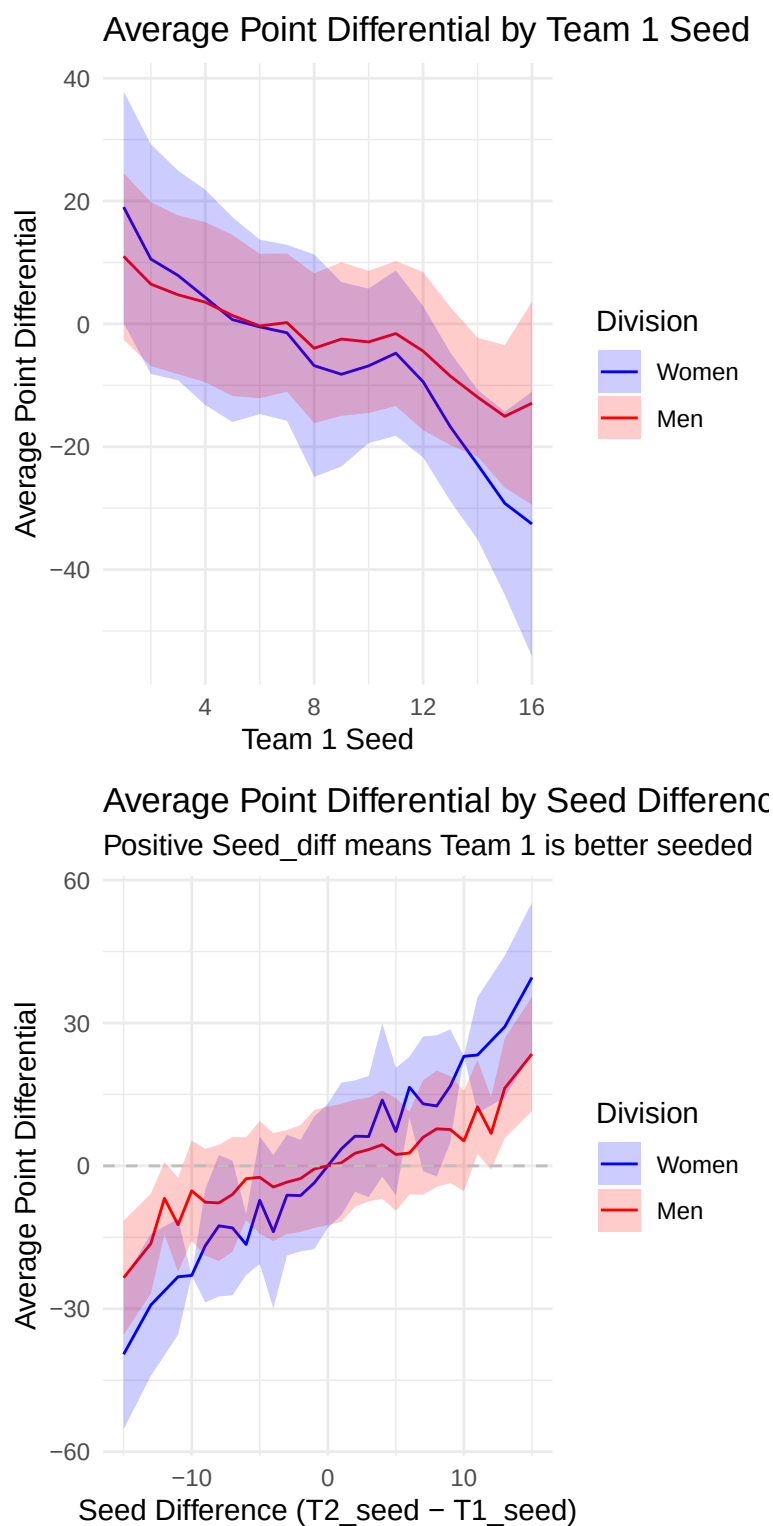


Figure 8. Left: Average point differential by Team 1 seed. Right: Average point differential by seed differential.

The second tier consists of simple season statistics computed from regular season game logs. For each team these include season win percentage, average points scored, average points allowed, and average point differential. Pairwise differentials between Team 1 and Team 2 are also computed for each of these statistics, providing the model with both absolute team quality signals and direct matchup comparison signals. These features capture overall team strength but do not account for the quality of opposition faced. A team that averaged 85 points per game against a weak conference schedule is not necessarily more dangerous than a team that averaged 75 points against a gauntlet of top 25 opponents, and the raw averages alone cannot make that distinction. This limitation motivates the third tier.

The third tier introduces strength of schedule adjusted box score profiles. For each of the fourteen raw box score statistics available in the Kaggle data, including field goals made and attempted, three point attempts, free throw attempts, offensive and defensive rebounds, assists, turnovers, steals, blocks, and personal fouls, seasonal averages are computed from two perspectives. The first perspective captures a team’s own offensive and defensive production, prefixed as `Off_` and `Def_` respectively. The second perspective captures the quality of opposition faced by averaging the seasonal profiles of every opponent a team played during the regular season, producing a set of `SOS_` prefixed features. A team whose offensive statistics were produced against consistently strong defensive opponents is credited more heavily than a team whose statistics were inflated by a weak schedule. All box score statistics are adjusted for overtime prior to averaging using the formula $\text{adjOT} = \frac{40+5 \times \text{NumOT}}{40}$, which scales each game’s statistics back to a standard 40

minute equivalent. This ensures that high scoring overtime games do not artificially inflate a team's seasonal averages.

The final component of the feature set is a custom Elo rating system developed to capture dynamic team strength across the full regular season. Unlike the static season averages described in the previous tiers, Elo ratings are updated after every game and reflect the cumulative result of a team's performance relative to its opponents throughout the season. This means a team that starts slowly but peaks heading into the tournament will have a higher Elo rating than its season averages alone would suggest, and conversely a team that fades late in the season will have a deflated rating relative to its win percentage.

The Elo system used here is adapted from the rating system originally developed for chess by Arpad Elo [10]. Each team begins the first season with an initial rating of 1500. After each regular season game the winner gains rating points and the loser loses an equal number, preserving the total rating mass in the system. The magnitude of the exchange depends on how surprising the result was relative to the pre-game expectations implied by the two teams' current ratings. The expected win probability for Team 1 is computed as:

$$(14) \quad E_1 = \frac{1}{1 + 10^{(R_2 - R_1)/400}}$$

where R_1 and R_2 are the current ratings of Team 1 and Team 2 respectively. The denominator of 400 is the width parameter that controls

how sensitive the expected probability is to rating differences. After the game resolves, ratings are updated as:

$$R_1 = R_1 + k(1 - E_1)$$

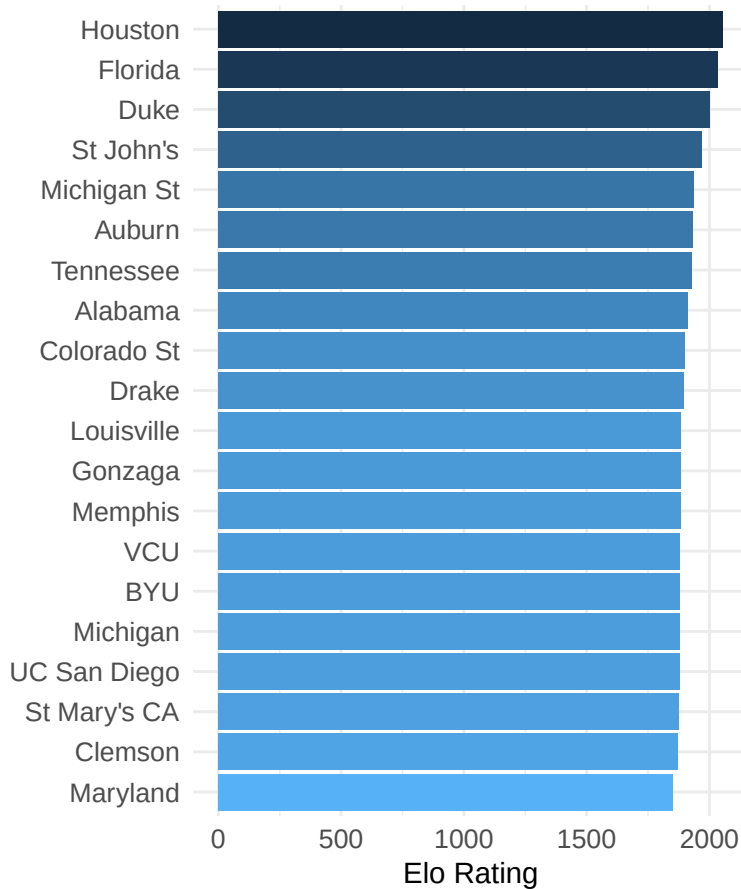
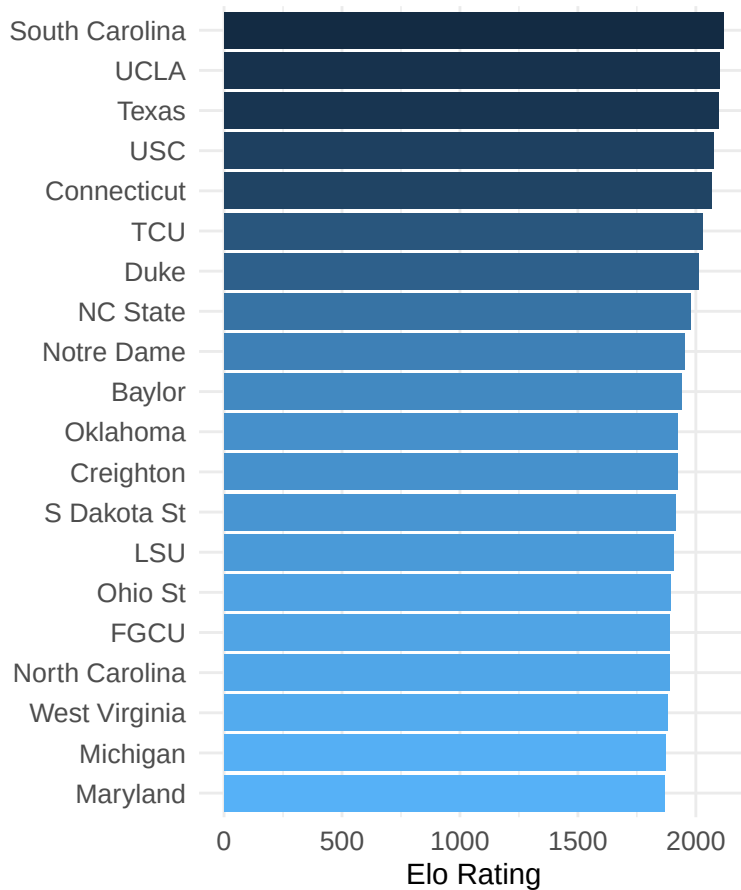
$$R_2 = R_2 + k(0 - E_2)$$

where $k = 64$ is the K-factor controlling the maximum possible rating change per game and $E_2 = 1 - E_1$. When a heavy favorite wins, E_1 is close to 1 and the rating change is small since the result was expected. When a heavy underdog wins, E_1 is close to 0 and the rating change is large. This property makes Elo particularly well suited for tournament prediction because late season upsets and dominant runs cause large rating movements that static averages cannot capture. A key design decision in this implementation is the between-season carry over. Rather than resetting every team to 1500 at the start of each new season, the end of season rating is partially carried forward according to:

$$(15) \quad R_{\text{new}} = 0.5 \times R_{\text{final}} + 750.$$

This means 50% of a team's historical rating is preserved into the next season while the other 50% regresses toward the mean. The motivation is that program quality is not purely year to year. A program like Duke or Connecticut that has been consistently elite for two decades should not start each season at the same rating as a mid-major program making its first

tournament appearance. The carry over allows the Elo system to accumulate this historical signal while the regression toward the mean prevents ratings from drifting arbitrarily far from the initial value over many seasons. Figures 9 and ?? illustrate the end of season Elo ratings for the top 20 men's and women's programs in 2025, reflecting both current season performance and accumulated historical quality. Each team's final end of season Elo rating is included in the feature set as `T1_Elo` and `T2_Elo`, along with their difference `Elo_diff`. Including all three again allows the model to learn both absolute and relative rating effects simultaneously.

Top 20 Men's Teams 2025**Top 20 Women's Teams 2025**

Data Preparation

The training data for the XGBoost model is constructed from all NCAA tournament games between 2003 and 2024, combining both the men’s and women’s divisions into a single dataset. The choice of 2003 as the starting point reflects the availability of detailed box score statistics in the Kaggle data, as earlier seasons contain only compact results without the possession level statistics needed to compute the strength of schedule adjusted features.

A critical design decision in the data construction is the symmetric duplication of each tournament game. For every game in the dataset, two rows are created: one where the winning team is assigned to the T1 position and one where the losing team is assigned to the T1 position. The binary outcome variable is set to 1 when T1 wins and 0 when T1 loses. This duplication serves two purposes. First, it allows the model to learn from both the winner’s and loser’s perspective in every matchup, meaning the model sees what a strong team’s feature profile looks like when it wins and what a weak team’s feature profile looks like when it loses. Second, it ensures that the model does not develop a positional bias toward the T1 slot, since every game appears from both orientations the model cannot exploit any systematic advantage of being labeled T1. This doubles the effective size of the training data, which is particularly valuable given that the total number of tournament games across all seasons is modest relative to the complexity of the feature set.

Once all features are computed, the training matrix is filtered to retain only rows where every feature is present using `complete.cases`. This step is included as a safeguard against missing values that could arise if a team appeared in the tournament without sufficient regular season history to

compute stable averages, or if the Elo system had not yet accumulated enough games to generate a meaningful rating. In practice no rows were removed, indicating that every tournament team in the dataset had sufficient regular season history to produce a complete feature profile across all seasons from 2003 to 2024. This is partly a consequence of the 2003 cutoff, which ensures that every season in the training data has a full year of regular season results available before the tournament begins.

The 2025 test set is constructed to exactly mirror the training data pipeline. Every feature engineering step, merge, and column ordering applied during training is repeated identically for the test set. This is a critical requirement for tree based models since XGBoost maps each input column to a specific feature index at prediction time and any mismatch between the training and test feature matrices will produce incorrect predictions. Season identifiers, team identifiers, and games played counts are excluded from both the training and test matrices for the same reason they are excluded from the logistic regression: they carry no generalizable predictive signal and would cause the model to memorize season specific patterns rather than learn relationships that transfer to future tournaments.

Model Implementation

With the training matrix assembled and the feature set finalized, the focus shifts to training the XGBoost model and selecting the configuration that produces the best out of sample performance. Two key decisions must be made before training can begin: the number of boosting rounds, which determines how many sequential trees are added to the ensemble, and the

hyperparameter values that control the complexity and regularization of each individual tree. Both decisions have a meaningful impact on model performance and neither can be determined from the training data alone without risking overfitting. To address this, the model uses five fold cross validation to guide both the round selection and the hyperparameter tuning process.

Cross validation is a resampling technique used to evaluate model performance on data that was not used during training. The core motivation is that evaluating a model on the same data it was trained on produces an overly optimistic estimate of performance, since the model has already seen and fit to that data. A naive solution is to hold out a fixed portion of the data as a test set, but with a small dataset this sacrifices training observations that could meaningfully improve the model. Cross validation addresses this trade off by rotating which portion of the data is held out across multiple iterations, allowing every observation to serve as both a training example and a validation example at different points. In k -fold cross validation the dataset is partitioned into k equally sized subsets called folds. The model is trained k times, each time using $k - 1$ folds as the training set and the last fold as the validation set [22]. The performance metric is recorded for each fold and the final cross validated score is the average across all k folds. This produces a more reliable estimate of out of sample performance than a single train test split because the result is averaged over multiple different held out subsets rather than depending on one particular partition of the data. Extensive research has gone into cross validation and general cross validation. For more information, we refer the reader to [21].

In this analysis five fold cross validation is applied to the full tournament training dataset to determine the optimal number of boosting rounds. At each fold the model is trained on 80% of the tournament games and evaluated on the held out 20%, with the average Brier Score across all five folds used as the selection criterion. Early stopping is applied with a patience of 50 rounds, meaning training halts automatically if the cross validated Brier Score does not improve for 50 consecutive rounds, preventing the model from overfitting by adding trees that contribute no meaningful reduction in error. The optimal number of rounds identified by cross validation is then used to train the final model on the complete training dataset. Five folds was chosen as a balance between bias and variance in the performance estimate. Using fewer folds, such as three, means each validation set is larger but the training set is smaller, which can underestimate the model's true performance on a full dataset. Using more folds, such as ten, produces a training set that more closely resembles the full dataset but increases computational cost and can produce high variance estimates when the dataset is small.

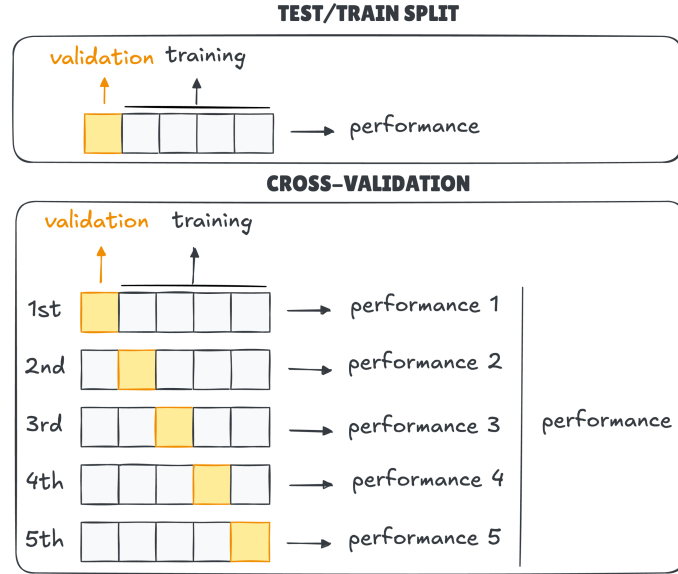


Figure 10. Example of Cross Validation and Regular Test/Train Split [19]

In addition to the number of boosting rounds, five hyperparameters were tuned to control the complexity and regularization of the model. Each parameter was tuned independently by fixing all others at their current best value and testing candidate values one at a time, selecting the value that produced the lowest cross validated Brier Score before moving to the next parameter. The learning rate **eta** controls how much each new tree contributes to the ensemble. A lower value of **eta** requires more boosting rounds to converge but produces a smoother and more regularized model. A value of **eta** = 0.01 was selected after testing values of 0.02 and 0.005, with 0.01 producing the lowest cross validated Brier Score. The **max_depth** parameter controls the maximum depth of each individual tree, limiting how many sequential splits a tree can make and therefore how complex the interactions it can capture. A value of **max_depth** = 5 was selected after

testing values of 4 and 6. The `subsample` parameter controls the fraction of training rows randomly sampled when building each tree, introducing stochasticity that reduces overfitting. A value of `subsample = 0.8` was selected, meaning each tree is built on a random 80% of the training data. Similarly, `colsample_bytree = 0.7` means each tree randomly samples 70% of the available features, further reducing the influence of any single predictor and encouraging the model to find robust signals across the full feature set. Finally, `min_child_weight = 4` sets the minimum number of training observations required in a leaf node before a split is made, preventing the model from creating highly specific splits that apply to only a handful of games. Together these five parameters define a model that is expressive enough to capture nonlinear tournament dynamics while remaining sufficiently regularized to generalize to unseen data.

Once the optimal hyperparameters and number of boosting rounds are determined through cross validation, the final model is trained on the complete training dataset using those configurations. Training on the full dataset rather than a held out split is justified here because the total number of tournament games across all seasons is relatively small, and withholding a portion of that data for a final validation set would meaningfully reduce the signal available during training. The cross validated Brier Score already provides a reliable estimate of out of sample performance, so a separate held out test set is not necessary to evaluate the model before deployment. The final model is then applied to the 2025 tournament games to generate win probabilities for each matchup, which are evaluated against the actual outcomes using the Brier Score.

Point Spread Models

same as above

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Here are my insights...

Comparison of models

2026 Competition

Strategies

Here are my insights...

Results

CONCLUSIONS

Summary

As a preliminary assessment...

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